

Searching

Data Structure(CSL202)

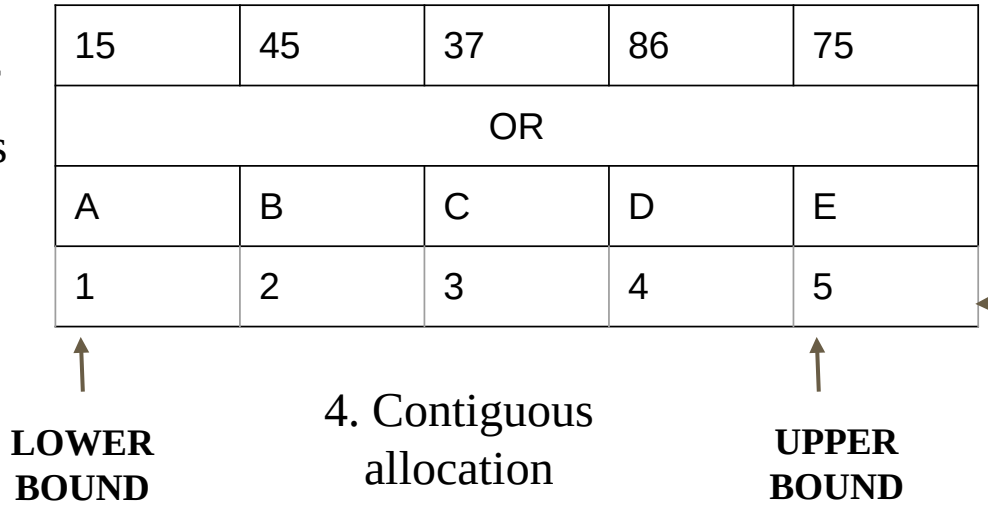
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Array

Array

1. Collection of homogeneous elements.



Length of array = UPPER BOUND - LOWER BOUND + 1

Accessing Elements of an Array(1-D

- Byte address of an Element

$$L(A[Index]) = BA + Size\ (Index - LB)$$

Where

Index = index of element

BA = Base Address of Array

Size = Storage size of one element in bytes.

LB = Lower bound of the index.

Accessing Elements of an Array(1-D

A[-5 ….. +5], Base address = 999, Size of element = 100 bytes. Find the location of A[2]?

Here

Index = 2, BA = 999, Size = 100 bytes, LB = -5

L(A[Index]) = BA + Size (Index - LB)

L(A[2]) = 999 + 100 * (2 - (-5))

L(A[2]) = 999 + 700

L(A[2]) = 1699

Accessing Elements of an Array(2-D

In Row-major Order:

- Byte address of an Element

$$L(A[i][j]) = BA + Size((i - LR) * N + (j - LC))$$

Where

i = row index of element, j = column index of element

BA = Base Address of Array

Size = Storage size of one element in bytes.

LR = Lower Limit of Row Index, LC = Lower Limit of Column Index

N = Number of column

Accessing Elements of an Array(2-D

Let A [1 …. 10] [1 ….. 15] be a two dimensional array of integer; Assuming that each integer takes one byte in memory, the array is stored in row-major order and the first element of the array is stored at location 100, what is the address of the element A[2][3]?

Here

i = 2, j = 3, BA = 100, Size = one byte, LR =1, LC = 1, N = 15

L(A[i][j]) = BA + Size ((i - LR) * N + (j - LC))

L(A[2][3]) = 100 + 1((2 - 1)* 15 + (3 - 1))

L(A[2][3]) = 117

Accessing Elements of an Array(2-D

In Column-major Order:

- Byte address of an Element

$$L(A[i][j]) = BA + Size((j - LC) * M + (i - LR))$$

Where

i = row index of element, j = column index of element

BA = Base Address of Array

Size = Storage size of one element in bytes.

LR = Lower Limit of Row Index, LC = Lower Limit of Column Index

M = Number of rows

Accessing Elements of an Array(2-D

Let A [1 …. 10] [1 ….. 15] be a two dimensional array of integer; Assuming that each integer takes one byte in memory, the array is stored in row-major order and the first element of the array is stored at location 100, what is the address of the element A[2][3]?

Here

i = 2, j = 3, BA = 100, Size = one byte, LR =1, LC = 1, N = 15

L(A[i][j]) = BA + Size ((j - LC) * M + (i - LR))

L(A[2][3]) = 100 + 1((3 - 1)* 10 + (2 - 1))

L(A[2][3]) = 121

Exercise

Consider the linear arrays A(15:87), B(-75:-8) and C(30).

(a) Find the number of elements in each array.

(b) Suppose Base(A) = 300 and w = 4 words per memory cell for A.

Find the address of A[34], A[58] and A[79].



Applications of Array

Calendar Days

AUGUST 2025

SUNDAY	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY
					1	2
3	4	5	6	7	8	9 Full moon
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

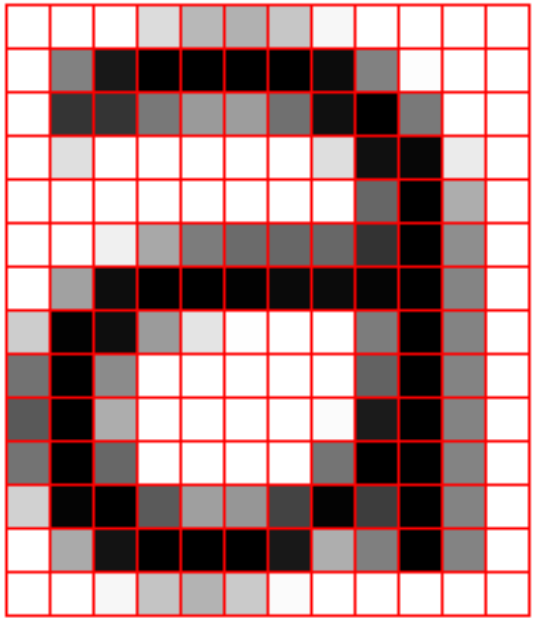
Daily Timetables / Schedules

Schedule for classes

Day	8:30-9:25	09:30-10:25	10:30-11:25	11:30-12:25	12:30-1:25	02:30-03:25	03:30 - 4:25	4:30 - 5:25
Monday	A	**N**				**O**		
		B	C	E	G	K	L	I
Tuesday	B	**V**				**W**		
		D	C	**X**	H	L	J	M
Wednesday	E	**P**				**Q**		
		A	D	F	G	K	M	I
Thursday	F	**R**				**S**		
		B	C	E	H	L	M	J
Friday	G	**T**				**U**		
		A	D	F	H	K	J	I

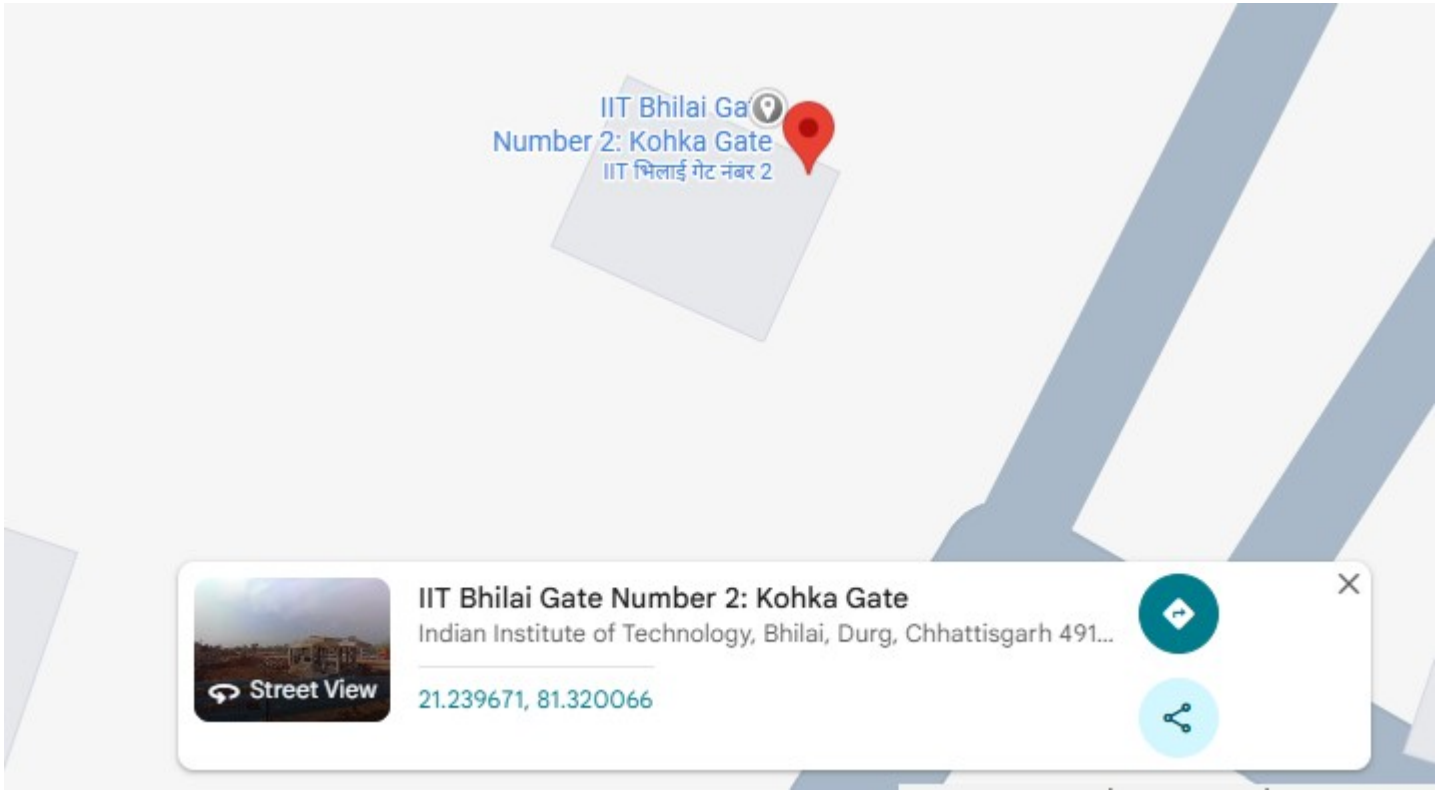
Pixel Values in Digital Images

a



1.0	1.0	1.0	0.9	0.6	0.6	0.6	1.0	1.0	1.0	1.0	1.0
1.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.5	1.0	1.0	1.0
1.0	0.2	0.2	0.5	0.6	0.6	0.5	0.0	0.0	0.5	1.0	1.0
1.0	0.9	1.0	1.0	1.0	1.0	1.0	0.9	0.0	0.0	0.9	1.0
1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.5	0.0	0.5	1.0
1.0	1.0	1.0	0.5	0.5	0.5	0.5	0.5	0.4	0.0	0.5	1.0
1.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	1.0
0.9	0.0	0.0	0.6	1.0	1.0	1.0	1.0	0.5	0.0	0.5	1.0
0.5	0.0	0.6	1.0	1.0	1.0	1.0	1.0	0.5	0.0	0.5	1.0
0.5	0.0	0.7	1.0	1.0	1.0	1.0	1.0	0.0	0.0	0.5	1.0
0.6	0.0	0.6	1.0	1.0	1.0	1.0	0.5	0.0	0.0	0.5	1.0
0.9	0.1	0.0	0.6	0.7	0.7	0.5	0.0	0.5	0.0	0.5	1.0
1.0	0.7	0.1	0.0	0.0	0.0	0.1	0.9	0.8	0.0	0.5	1.0
1.0	1.0	1.0	0.8	0.8	0.9	1.0	1.0	1.0	1.0	1.0	1.0

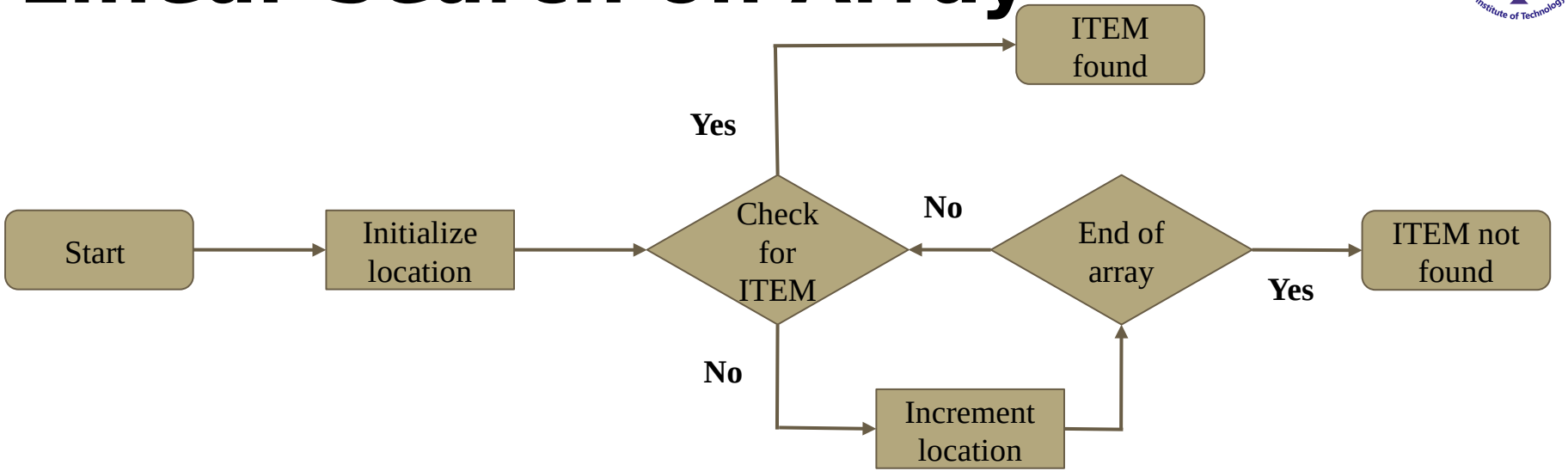
GPS Coordinates





Linear Search on Array

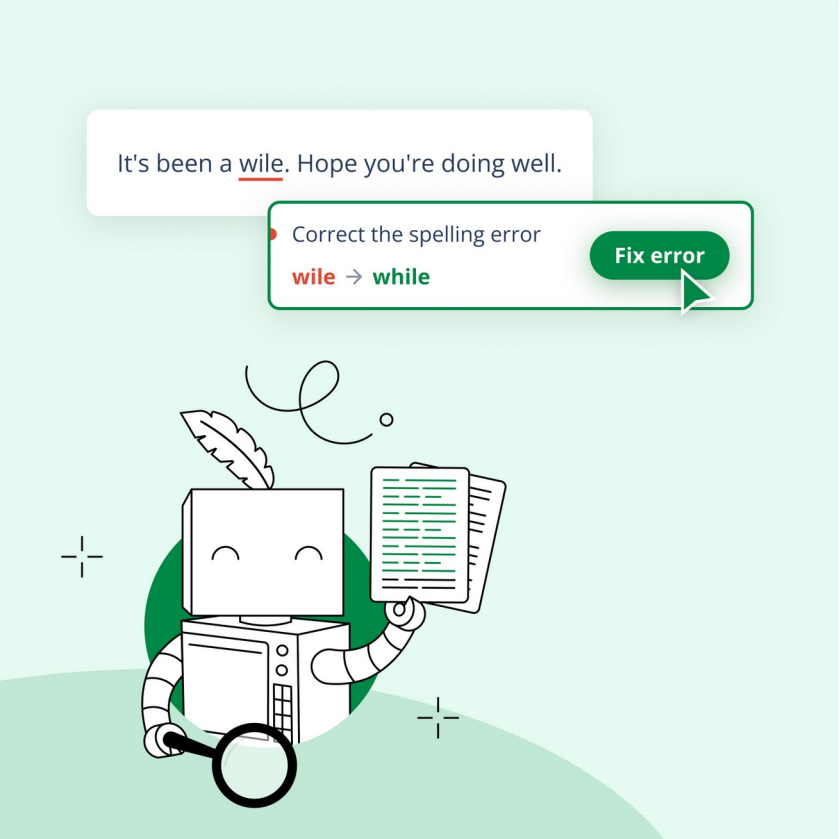
Linear Search on Array



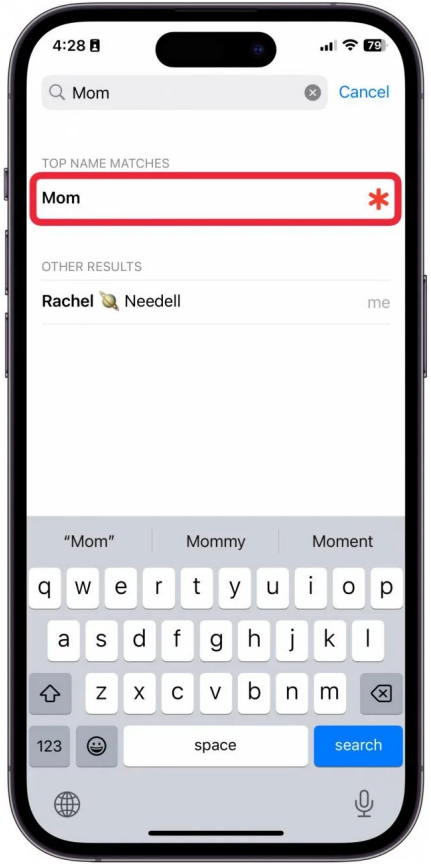
Time Complexity:

- **Best Case:** ITEM is at the first position (O(1))
- **Worst Case:** ITEM is not in the array, or it's at the last position (O(n)).
- **Average Case:** ITEM is anywhere in the array(O(n)).

Spell Checkers



Search Contacts on Your Phone



Linear Search

ITEM = 62

9	15	28	33	48	62	76	90	108	132	159	220	357

A[LOC] ≠ ITEM

9	15	28	33	48	62	76	90	108	132	159	220	357

A[LOC] ≠ ITEM

9	15	28	33	48	62	76	90	108	132	159	220	357

A[LOC] ≠ ITEM

9	15	28	33	48	62	76	90	108	132	159	220	357

A[LOC] ≠ ITEM

Linear Search

ITEM = 62

9	15	28	33	48	62	76	90	108	132	159	220	357

A[LOC] ≠ ITEM

9	15	28	33	48	62	76	90	108	132	159	220	357

A[LOC] = ITEM

ITEM found search successful.

6 Comparisons

Exercise

Consider the alphabetized linear array SUBJECT

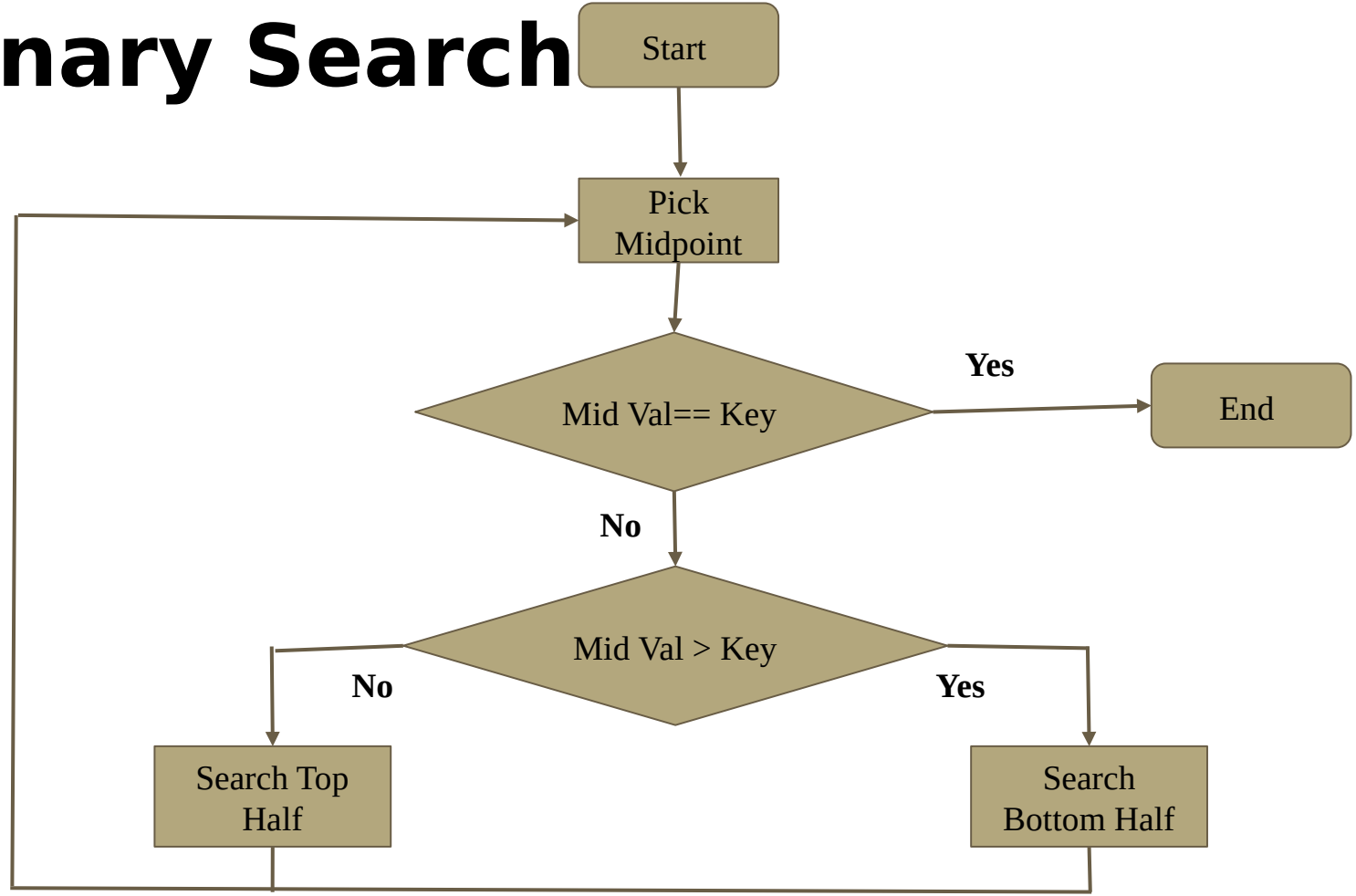
(a) How many comparisons are used to locate DSA, DM and NM?

1	Programming
2	Linear Algebra
3	DSA
4	DBMS
5	DM
6	Digital Logic
7	Fundamental
8	NM
9	Microprocessor
10	Software Engineering



Binary Search on Array

Binary Search



Binary Search

```
void BINARY(int DATA[], int LB, int UB, int Key, int *LOC)
{
        int BEG = LB;
                int END = UB;
                int MID = (BEG + END) / 2;
                while (BEG <= END && DATA[MID] != Key)
        {
                                if (Key< DATA[MID])
                {
                                END = MID - 1;
                                }
                else
                {
                                BEG = MID + 1;
                                }
                                MID = (BEG + END) / 2; // Update MID
                }
```

```
        if (DATA[MID] == Key)
        {
                                *LOC = MID; //
Found
                }
        else
        {
                                *LOC = 0;   // Not
found
                }
}
```

- **Best Case:** Target element is found at the middle in first try (O(1)).
- **Worst Case:** Key not in array or at one end (O(log n)).
- **Average Case:** Target may be anywhere(O(log n)).

Binary Search

Key = 62

9	15	28	33	48	62	76	90	108	132	159	220	357

BEG = 1 **MID = 7** **END = 13**

62 < 76

9	15	28	33	48	62	76	90	108	132	159	220	357

BEG = 1 **MID = 3** **END = 6**

62 > 28

9	15	28	33	48	62	76	90	108	132	159	220	357

BEG = 4 **MID = 5** **END = 6**

62 > 48

Binary Search

9	15	28	33	48	62	76	90	108	132	159	220	357

BEG = 6 **MID = 6** **END = 6**

A[MID] = Key

Key found, search successful.

4 Comparisons

Exercise

Given an array arr = {12, 13, 35, 78, 56} and key = 78; How many iterations are done until the element is found using Binary search?